

Thong (Tom) Huynh

 ThongHHuynh |  Thong Huynh |  Portfolio |  thong.huynh@ontariotechu.net

EDUCATION & AWARDS

Ontario Tech University	B.Eng., Mechatronics Engineering (GPA: 4.25/4.3)	2024–2028
Competitions	International Youth Robot Competition 2019 (1st place), MakeX (Innovative Design), FIRST Robotics	
Awards	Undergraduate Research Award, STAR Award, Merit Award, President’s List	

SKILLS

CAD and Design:	SolidWorks Certified Professional , Siemens NX, 3D-Printing, GD&T, DFM/DFA.
Robotics Hardware:	NVIDIA Jetson Orin Nano/AGX, Raspberry Pi, LiDAR, IMUs, sensors, actuators.
Robotics and Simulation:	ROS 2, Gazebo, NVIDIA Isaac Sim/Lab, Nav2.
Programming and Vision:	Linux, Python, C++, Git, OpenCV, PyTorch, YOLO, RF-DETR, DINO/DINOv2.

EXPERIENCE

Robotics Software Engineer | ABI Ltd.

May 2026 – Present

- Developed end-to-end digital twins of industrial robotic workcells in **NVIDIA Isaac Sim** to accelerate system validation, reducing on-site commissioning time by 30%.
- Architected an in-house, two-stage YOLO with CNN head for a bakery inspection pipeline (1M+ pieces/day), achieving < 100 ms inference latency and cutting \$60K in outsourcing costs.
- Engineered an internal computer vision tooling suite deployed at 2 customer sites and adopted company-wide across the engineering team, improving model evaluation and deployment workflows.
- Supported the integration of 6-DOF ABB robotic arms for high-speed automated bread-scoring systems.

Research Assistant | MARS Lab

May 2025 – May 2026

- Developed a hybrid multimodal walking-and-flying robotic platform and designed 10+ 3D-printed mechanical components (carbon fiber and SLS nylon) to optimize structural rigidity and payload capacity by 20%.
- Formulated forward/inverse kinematics for quadruped locomotion and built an **end-to-end Gazebo simulation pipeline** for dynamic gait and motion validation.
- Implemented OctoMap-based 3D mapping, navigation, and obstacle avoidance via MAVROS.

Design Team Lead | Ontario Tech RoboMaster

September 2024 – Present

- Directed the mechanical design of a 500+ component competition robot in SolidWorks with **full DFM/DFA compliance** for CNC machining, sheet metal, and 3D printing.
- Engineered a 2-DOF turret system with belt and herringbone gear, improved the accuracy and speed of the robot by 27%.
- Performed **failure analysis and FEA**, managed CAD version control, and standardized team processes to ensure consistency across subsystems.
- Led a team of 6+ team members in mechanical design, **GD&T**, engineering drawings, and robotics design principles.

PROJECTS

SensQ – Autonomous Mobile Robot Platform

January 2026 – Present

- Designed and 3D-printed an autonomous mobile robot platform, achieving speeds of 1 m/s with sub-2mm repeatability in point-to-point navigation tasks.
- Simulated system performance in Gazebo and integrated end-to-end physical hardware with ROS 2.
- Implemented Nav2 and SLAM Toolbox navigation, custom A* and Boustrophedon coverage planning, and AprilTag-based docking.

Bolt Eye – Automated Inspection System

August 2025 – Present

- Built and integrated a 1 m/s conveyor with encoder tracking and synchronized vision acquisition for real-time inspection.
- Developed a real-time defect detection and classification pipeline using YOLOv11 and DINOv2, achieving sub-100 ms inference latency.
- Detected and classified surface defects as small as 2 mm on an active production-line conveyor operating at 1 m/s.

6-DOF Robotic Arm

May 2024 – January 2026

- Designed a 6-DOF robotic arm with custom 3D-printed structural linkages, calculating preload compensation and torque analysis for each joint, increasing the payload of the arm by 15%, up to 2kg.
- Customized kinematics calculation with L-BFGS-B path planning for pick and place tasks with 2mm accuracy.
- Implemented with Python, MoveIt, and RViz tools for motion visualization and validation.